

1

a

$$\hat{M}_{MLE} = x_n$$

b

Given from the announcement:

$$E(\hat{M}_{MLE}) = \frac{n}{n+1}(M+1)$$

This is a biased estimator, as $\hat{M}_{ME} \leq M$ and will underestimate M .

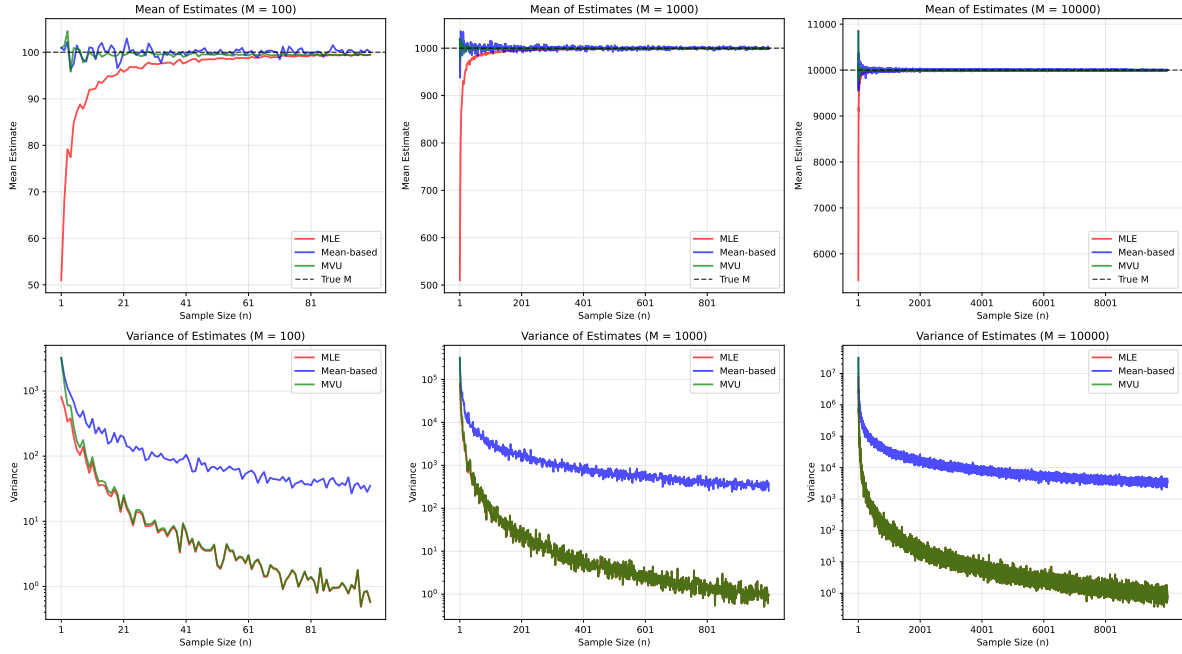
$$E(\hat{M}_{MEAN}) = M$$

This is an unbiased estimator as the expected value of M_{MEAN} is equal to the value it is trying to estimate (M)

$$E(\hat{M}_{MVU}) = M$$

This is an unbiased estimator as the expected value of M_{MVU} is equal to the value it is trying to estimate (M)

C

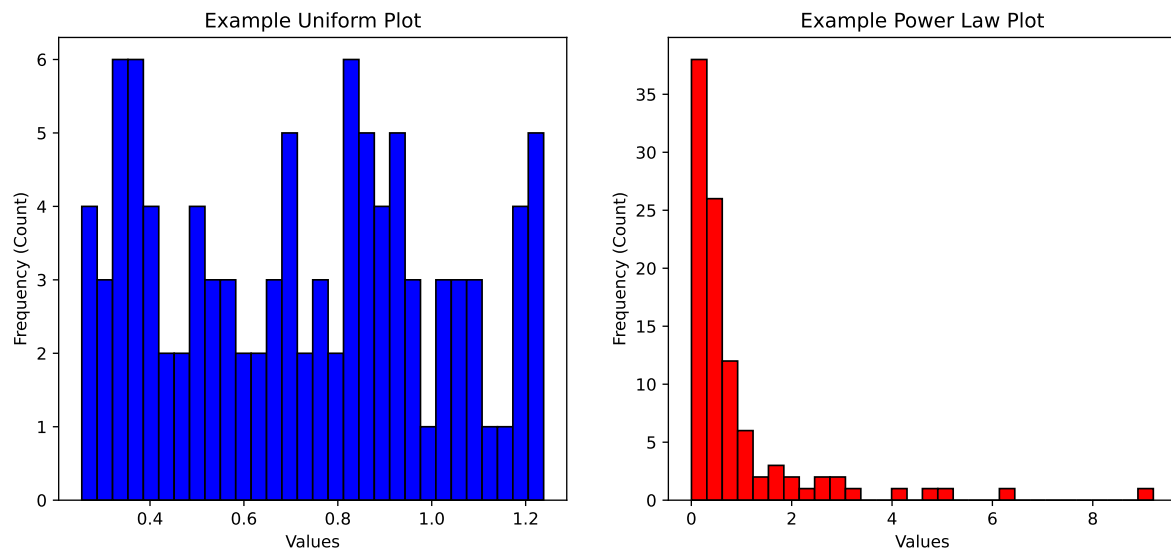


According to the Mean of estimates graph it is clear to see that the $\hat{M}_{MEAN}$ and $\hat{M}_{MVU}$ estimates are unbiased estimators of M as they their expected value is M . On the other hand, $\hat{M}_{MLE}$ is evidently an unbiased estimator as it falls significantly below M for smaller values of n . For sufficiently large n , the three estimators all approach M which is good.

Their variances on the other hand tell a different story. The $\hat{M}_{MEAN}$ estimator has an extremely volatile variance many orders of magnitude above both $\hat{M}_{MLE}$ and $\hat{M}_{MVU}$. This demonstrates that the $\hat{M}_{MVU}$ estimator should be the best estimator for M as the variance is tied for smallest and is an unbiased estimator of M .

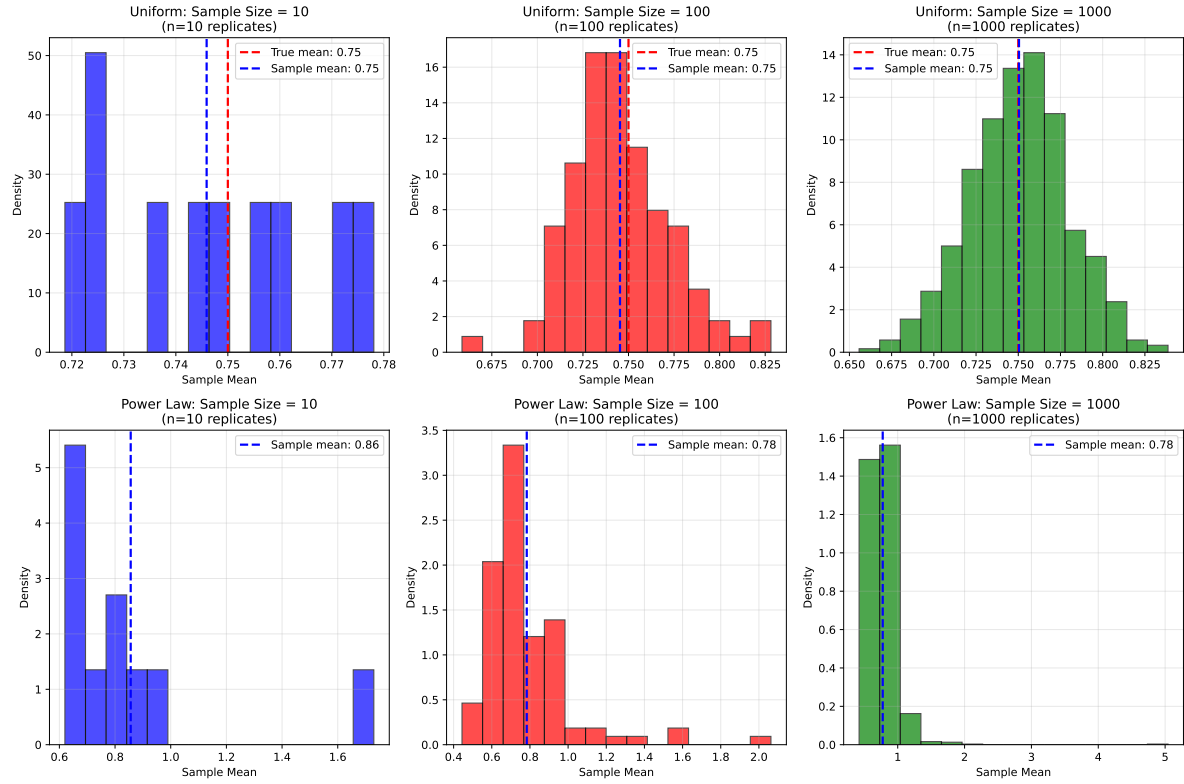
2

a

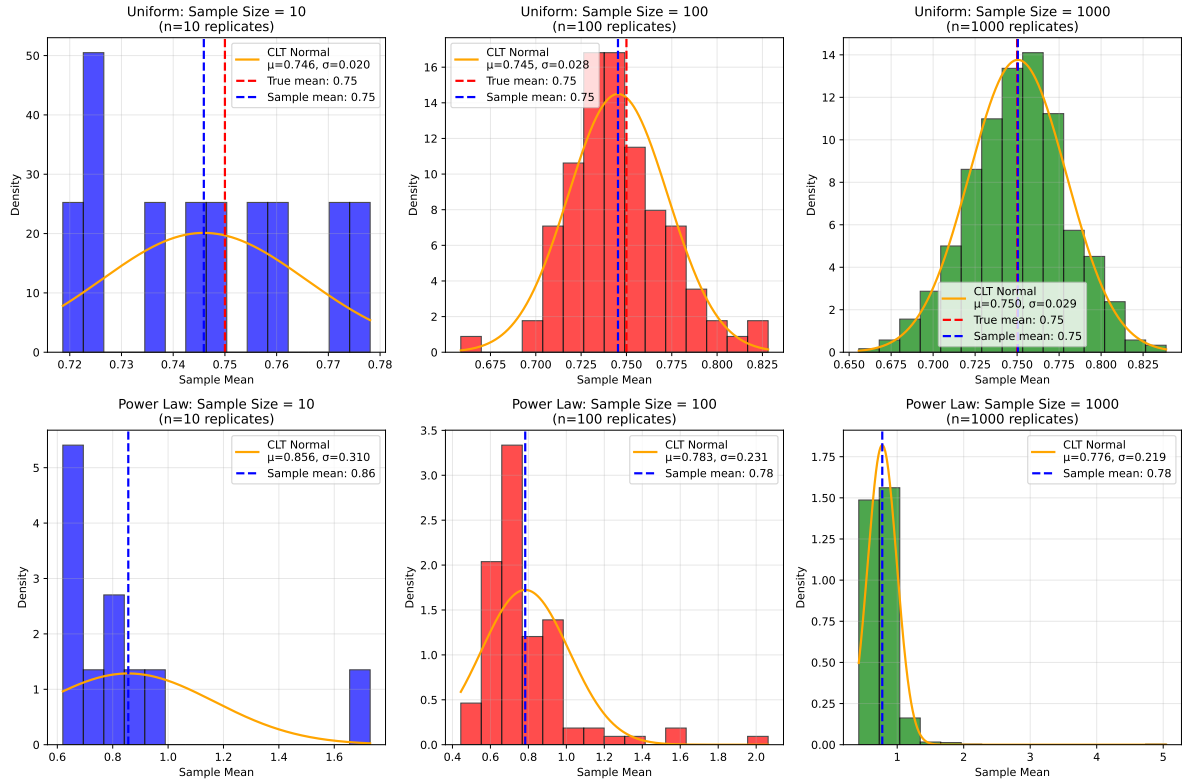


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b



c



d

Uniform

Samples	KS Statistic	P-Value
10	0.178	0.855
100	0.079	0.541
1000	0.017	0.929

Power Law

Samples	KS Statistic	P-Value
10	0.268	0.397
100	0.160	0.011

Samples	KS Statistic	P-Value
1000	0.126	2.28×10^{-14}

First thing to do was to standardize the distributions to attempt to compare them to $N(0, 1)$. For both the uniform distribution and the power law distribution, as the number of samples increased then the KS Statistic decreased. This is as expected because CLT refers to as samples increase then the distribution is more normal. For uniform, all the p-values will fail to reject the null hypothesis (under any reasonable α), this means that the samples come from the normal distribution. However this isn't the case for the power law distribution. For $\alpha = 0.05$, the power law samples only fail to reject when $n = 10$, thus for $n = 100, n = 1000$ we reject the null hypothesis and say that the means do not follow the normal distribution. While this may seem off at first, according to the slides, the power law distribution is very slow to converge thus these outcomes make sense.

For the power of the KS-test as n increases, it is evident that the p-value converges to either 0 or 1 as n increases. This is intuitive as the number of samples increase the ks-test becomes more certain about the distribution matching to a normal distribution.

3

a

The number of US cars is 249 and the number of Japanese cars is 79 in this dataset.

b

Let μ_{US} be the population mean mileage for US cars and μ_{JP} be the population mean mileage for Japanese cars.

$$H_0 : \mu_{US} = \mu_{JP} \text{ and } H_1 : \mu_{US} < \mu_{JP}$$

c

To test our hypothesis we can use a two sample t-test. The test statistic is:

$$t = \frac{\bar{x}_{JP} - \bar{x}_{US}}{\sqrt{\frac{s_{JP}^2}{n_{JP}} + \frac{s_{US}^2}{n_{US}}}}$$

But first we would check the normality and the variances of the two car samples to see if this is the right test. When checking, the normality, we find that the US cars are not normally distributed. However using the central limit theorem (both samples are larger than 30 and are samples of sample means), we can say that the sample mean approaches a normal distribution so we will continue with our tests.

d

The value of the test statistic is 12.946, resulting in a p-value of 7.854×10^{-26} .

e

Since the p-value is smaller than any reasonable α we reject the null hypothesis and there is enough statistical evidence to support the salesman's claim.